

pyLEEM: A Python package for dynamic Low-Energy Electron Microscopy data analysis

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Summary

pyLEEM is a Python package designed to streamline the analysis of Low-Energy Electron Microscopy (LEEM) experimental data. The package provides a unified framework for reading raw data files, performing energy calibrations, extracting spectroscopy profiles, and conducting domain-specific analyses, including X-ray photoelectron spectroscopy (XPS) stitching, X-ray absorption spectroscopy (XAS), secondary electron emission spectroscopy (SEES) energy correction, and diffuse electron scattering pattern (DESP) characterization. Through a modular and extensible core architecture combined with batch processing capabilities, pyLEEM enables researchers to develop reproducible, automated analysis workflows for large-scale datasets.

Statement of Need

LEEM is a full-field, surface-sensitive electron imaging technique that employs low-energy electrons in the 1 – 100 eV range to study surface phenomena at nanometer-scale lateral resolution and millisecond temporal resolution. The combination of high lateral and temporal resolution with surface sensitivity makes it ideally suited for investigations of surface morphology, chemical dynamics, and in situ material growth (Bauer, 2020; Patel, Wang, et al., 2025; Rogge et al., 2015; Tromp, 2000). In addition to imaging, LEEM instruments can acquire energy-dependent spectroscopic measurements and reciprocal-space diffraction patterns, and can be integrated with photoemission electron microscopy (PEEM) to enable soft X-ray-based chemical analysis (Le et al., 2025; Mandziak et al., 2018; Sadowski, 2020). Together, these capabilities make LEEM a uniquely versatile tool for quantitative, time-resolved surface science, capable of correlating nanoscale structure, electronic properties, and dynamics within a single sample.

These dynamic measurements generate large, multidimensional datasets that require complex instrument calibration, drift and distortion correction, and extraction and stitching of spectroscopic data (De Jong et al., 2020). However, current analysis workflows typically rely on manual feature extraction in ImageJ (Schneider et al., 2012) or custom scripts, manual post-processing (e.g., energy-range calibration and stitching across the entire measurement stack), and manual calculations performed without using the measurement metadata. These practices are time-consuming, error-prone, and lack reproducibility. pyLEEM addresses these challenges by providing a modular and extensible Python framework with standardized input/output, image and metadata extraction, and support for domain-specific analysis and post-processing.

State of the field

Existing tools developed for LEEM analysis remain limited and are often highly customized and domain-specific (De Jong et al., 2020; Grady et al., 2018). As a result, many LEEM datasets are still analyzed using general-purpose image analysis tools such as ImageJ (Schneider et al., 2012) for image and profile extraction, along with non-trivial custom scripts for data analysis, post-processing, and conversion into target formats for downstream domain-specific tools. In addition, existing LEEM-related tools and ad hoc workflows do not provide the backend infrastructure needed for automated analysis, large dynamic datasets, or user-defined analysis modules. Therefore, we developed pyLEEM as a new package rather than extending existing projects.

Software design

pyLEEM is a modular, extensible Python package for LEEM-family data analysis. The key design trade-off was to create a reusable backend API, which adds upfront abstraction but makes pyLEEM easier to test, maintain, extend, and use in reproducible headless workflows. The package separates the workflow into three main layers: data input/output and metadata extraction, image and profile processing, and domain-specific analysis. This separation supports long-term maintainability by allowing input/output, calibration, image processing, and analysis operations to be developed, tested, and updated independently. The extensible structure also allows users and facility staff to implement custom workflows for new file formats, image-processing routines, and domain-specific analysis needs. The core pyLEEM library is intentionally separated from the graphical user interface, so that analysis routines remain readable, testable, and usable outside interactive sessions. This backend-first design enables large-scale batch processing through scripts, notebooks, and automated workflows.

A second design goal is workflow readability and reproducibility. Domain-specific workflows can be saved as a human-readable configuration file, either as an analysis template or to store analysis results. The design allows facility staff and tool managers to distribute validated analysis workflows directly to users, while users can reproduce the same sequence of operations on related datasets or future experiments.

Research impact statement

pyLEEM has supported both user-facility data analysis and method development. For the core developers, the package enables rapid development, testing, and validation of new workflows and analysis algorithms, including DESP calibration and characterization. More broadly, pyLEEM was developed at the Center for Functional Nanomaterials (CFN), a U.S. Department of Energy Office of Science user facility at Brookhaven National Laboratory. As a user facility, the CFN serves a broad community of visiting researchers, and pyLEEM has supported and continues to support users' data analysis at both the CFN and the National Synchrotron Light Source II (NSLS-II) (Patel, Wang, et al., 2025; Patel, Chen, et al., 2025).

AI usage disclosure

Generative AI tools were not used in the software creation, documentation, and paper authoring for pyLEEM. GPT-5.5 was used solely to review the existing codebase, docstrings, and comments for errors, clarity, and consistency. The authors made all core design decisions and reviewed, edited, and validated every change, and take full responsibility for the accuracy and originality of the submitted materials.

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